

Design Calculation Sheet - VBF-T8R2-CALC-01 (t8_r2, final design V_G)

1 inputs

Parameter	Value	Unit	Source
Electrode area	0.1294	m2	cell/r3_VG_energy.json
Thicknesses pos/neg/sep (um)	60 / 67.62 / 9	um	parameter set (Chen2020 base + cell/params_r3_VG.json overrides)
CC 8 Al / 6 Cu (um)	8 / 6	um	parameter set (Chen2020 base + cell/params_r3_VG.json overrides)
Porosities pos/neg/sep	0.335 / 0.25 / 0.47	-	parameter set
Nominal capacity / I_1C	5.0000	Ah / A	parameter set

2 capacity_energy

Quantity	Value	Unit
1C capacity (DFN)	5.0351	Ah
5C capacity (DFN)	4.8492	Ah
5C retention	0.9631	-
Discharge energy	18.2292	Wh
Midpoint voltage	3.9029	V
DCR (curve-derived)	0.0027	ohm

3 energy_density

Quantity	Value	Unit
Mass (contract caliber)	0.0377	kg
Energy density (gravimetric)	483.3667	Wh/kg
Energy density (volumetric)	935.2928	Wh/L
Peak power density (theoretical)	40422.2857	W/kg

4 np_mass

Quantity	Value	Unit
N/P (mechanical)	0.6674	-
Pos coating	16.8422	g
Neg coating	10.8741	g
Al CC	2.7951	g
Cu CC	6.9567	g
Separator	0.2450	g
Electrolyte (annotated)	6.4030	g

5 process

Parameter	Value	Unit
Pos areal density	130.1538	g/m2
Neg areal density	84.0336	g/m2
Pos compaction density	2.1692	g/cm3
Neg compaction density	1.2428	g/cm3
Electrolyte fill	6.4030	g